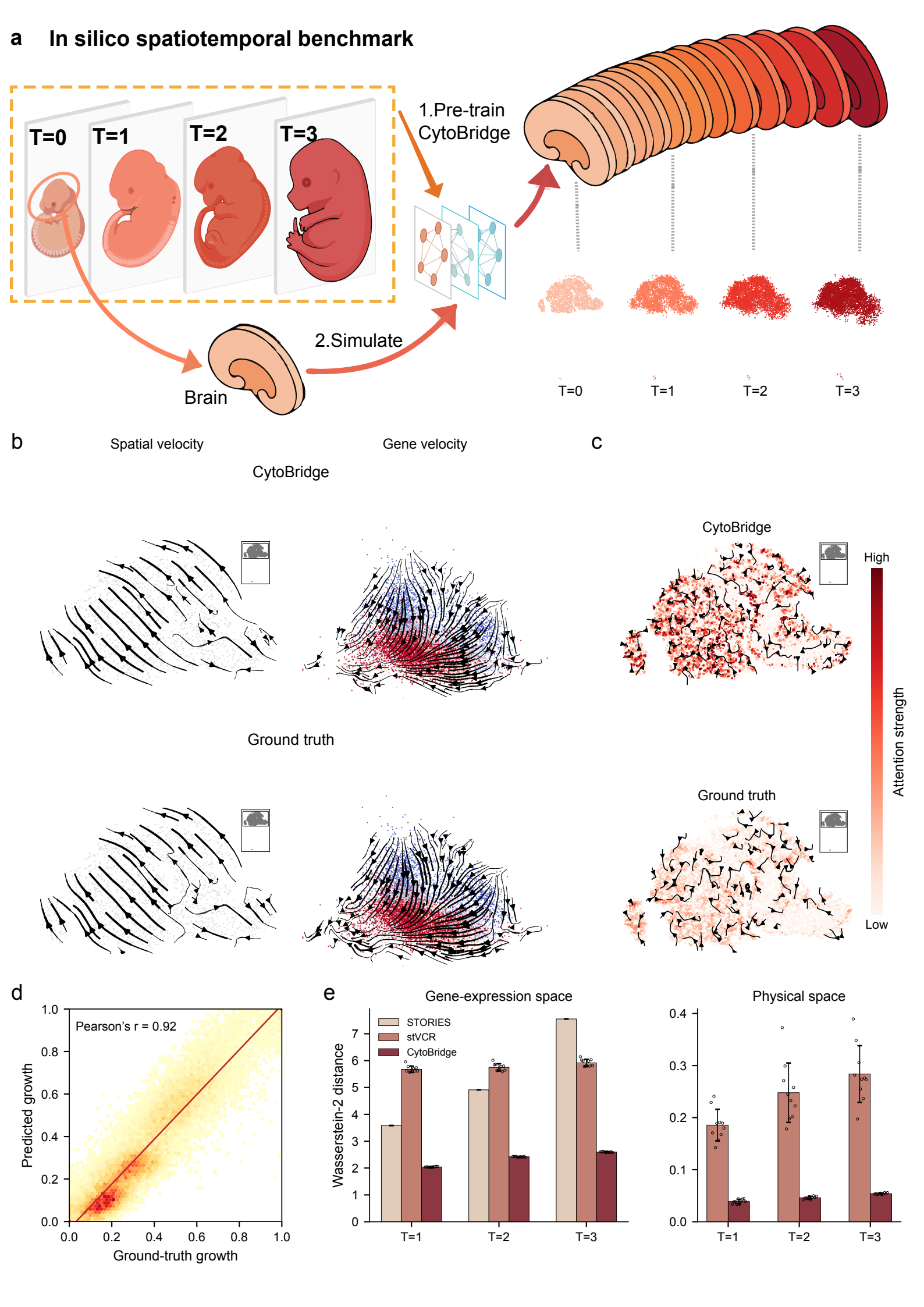

a
In silico spatiotemporal benchmark
T=0
T=1
T=2
T=3
1.Pre-train CytoBridge
2.Simulate
Brain
T=0
T=1
T=2
T=3
b
Spatial velocity
Gene velocity
CytoBridge
Ground truth
c
CytoBridge
Ground truth
High
Low
Attention strength
d
Pearson's r = 0.92
Predicted growth
Ground-truth growth
e
Gene-expression space
Physical space
STORIES
stVCR
CytoBridge
Wasserstein-2 distance
T=1
T=2
T=3